

Z-Class BB8 — Operator Card (v9.1)

Live PID Tuning & Calibration — PS3 Move Navigation Controllers

Controllers: Drive = Right controller (lx/ly, D-pad, PS, X/O), Dome = Left controller (lx/ly, D-pad, PS)

Enter Pitch Tuning: Hold **PS + UP** (Drive) for 3s → Pitch PID (KP → KD → KI).

Enter Roll (S2S) Tuning: Hold **PS + RIGHT** (Drive) for 3s → Roll PID (Pk2 → Dk2 → Ik2).

Cancel Tuning: Hold **PS + DOWN** (Drive) for 3s → Restore persisted gains, neutralize outputs.

During tuning: Joysticks, sounds, dome, flywheel are disabled. Use D-pad **UP/DOWN + X** (store/advance).

Pitch (Drive) PID Stages

Stage 1 — KP	Start KP=45.0, KI=0, KD=0; autoBalance OFF to see pure P. UP/DOWN: step ±1.0 . X: store KP with 25% safety reduction , persist, advance to KD.
Stage 2 — KD	autoBalance ON to observe damping. UP/DOWN: step ±1.0 . X: store KD, persist, advance to KI.
Stage 3 — KI	UP/DOWN: step ±0.1 . X: store KI, persist, exit tuning.

Roll (S2S) PID Stages

Stage 1 — Pk2	Start Pk2=45.0, Ik2=0, Dk2=0. UP/DOWN: step ±1.0 . X: store Pk2 with 25% reduction , persist, advance to Dk2.
Stage 2 — Dk2	S2S autobalance active for lateral damping observation. UP/DOWN: step ±1.0 . X: store Dk2, persist, advance to Ik2.
Stage 3 — Ik2	UP/DOWN: step ±0.1 . X: store Ik2, persist, exit tuning.

Calibration (3sec holds)

- **SAVE:** Hold **UP+UP** (both controllers) → Saves **potOffset**, **pitchOffset**, **rollOffset** to Preferences.
- **RESET:** Hold **DOWN+DOWN** (both) → Zeros offsets and persists.

Keys used: potOffset / pitchOffset / rollOffset.

Persistence & Cancel

- Stored gains: *kp_pitch*, *kd_pitch*, *ki_pitch*, *pk2*, *dk2*, *ik2* in Preferences.
- **PS+DOWN** cancel restores persisted gains, disables autobalance, and neutralizes drive/S2S PWM immediately.

Recommended Debug Combos

- Calibration: *DEBUG_CALIBRATION* + *DEBUG_COMPACT*
- PID Tuning: *DEBUG_S2S_MODE* + *DEBUG_MOTOR_OUTPUT* + *DEBUG_COMPACT*
- Controller: *DEBUG_JOYSTICK* + *DEBUG_COMPACT*